

Week 1

Random experiment: Activity, process, or experiment whose outcome(result) is uncertain.

Events: Collection(subset) of outcomes contained in the sample space.

Example 1: Two students living in university residence are selected and asked if they purchased a meal plan (P) or cook for themselves(C).

1, What is the random experiment?

Record response from both students.

2, What is the sample space?

$\Omega = ((P,P),(P,C),(C,P),(C,C))$

3, Give an example of an outcome?

(P,C): first student purchased a meal plan and second student cooks for themselves.

4, Give an example of an event?

A = at least one student cooks for themselves.

Set operations:

An event is a set.

Relationships and results from set theory can be used to construct new events from existing ones.

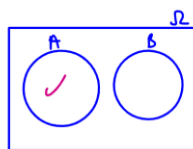
- The **union** of two events A and B, denoted by $A \cup B$, is the event containing all outcomes that are either in A or B or in both events.
- The **intersection** of two events A and B, denoted by $A \cap B$ (or simply AB), is the event consisting of all outcomes that are in both A and B.
- The **complement** of an event A, denoted by A^c , is the set of all elements of Ω that are not in A.

Independent: the occurrence of one event does not alter the chance that another event occurs.

$X \perp Y$

Dependent: the occurrence of one event alters the chance that the other event occurs.

Mutually exclusive/disjoint: if event A occurs, event B cannot occur.



also a special case of dependent

- Mutually independence vs pairwise independence:

$$P(X \cap Y \cap Z) = P(X)P(Y)P(Z) \quad \text{vs} \quad P(X \cap Y) = P(X)P(Y), P(Y \cap Z) = P(Y)P(Z), P(X \cap Z) = P(X)P(Z)$$

Mutual independence $\Rightarrow$ Pairwise independence

Venn diagram:

1, visually represent the sample space and events

2, To construct a Venn diagram:

Draw a rectangle whose interior represents the sample space Ω .

Any event is represented as the interior of a circle contained within Ω (circles may

overlap)

DeMorgan's laws

state that the complement of a union is an intersection, and the complement of an intersection is

For two events A and B :

- $(A \cup B)^c = A^c \cap B^c$
 - $(A \cap B)^c = A^c \cup B^c$
- a union.

Inclusion-exclusion:

$$P(X \cup Y \cup Z) = P(X) + P(Y) + P(Z) - P(X \cap Y) - P(X \cap Z) - P(Y \cap Z) + P(X \cap Y \cap Z)$$

Week 3

Multiplication Principle

- Fundamental Principle of Counting (FPC): a process has many steps, each step has $n_1, n_2, \dots, n_k$ possibility, total = $n_1 \times n_2 \times \dots \times n_k$
- **Permutation**: order matters, from n distinct elements selected k
 1. $P(n, k) = \frac{n!}{(n-k)!}$ 从 n 个不同元素中选 k 个排
 2. $n!$ n 个人有几种排法
 3. $\frac{n!}{n_1! n_2! \dots}$ 有重复元素的排列 $n_1 + n_2 + \dots = n$ e.g. 4 个元素中有 3 个 A 和 1 个 B, $\frac{4!}{3!1!}$
- **Combination**: unordered subsets, from n distinct elements selected k
$$C(n, k) = (\text{from } n \text{ choose } k) = \frac{n!}{k!(n-k)!}$$

9.17.2025

Conditional Probability

定义

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

若 independent, 则

$$P(A \cap B) = P(A)P(B)$$

且

$$P(A|B) = P(A)$$

Property of CDF:

- Nondecreasing Function
- Limits at Infinity
- Right continuous

Geo(p):

Property: Memoryless

因为 geometric distribution:

$$P(X > m) = (1 - p)^m$$

所以:

$$P(X > n + k | X > k) = \frac{P(X > n + k)}{P(X > k)}$$

代入:

$$= \frac{(1 - p)^{n+k}}{(1 - p)^k}$$

指数相减:

$$= (1 - p)^n$$

1. Discrete: PMF ($\rightarrow E(x)$, $\text{Var}(x)$ $\rightarrow$ characteristics of a RV), CDF, Percentile
2. Continuous

To represent the probability behaviour of a random variable: PMF probability mass function & CDF cumulative distribution function $F(x) \rightarrow$ the probability that the observed value of X will be at most x

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y).$$

The graph of CDF will be a non-decreasing step-function, right continuous, $0 \rightarrow 1$ (x from $-\infty$ to ∞)

$$P(a \leq X \leq b) = F(b) - F(a^-)$$

Percentiles and Quantiles

The p th percentile of a distribution is the value x_p such that: $F(x_p) = p/100$, $0 < p < 100$

The q th quantile is the value x_q : $F(x_q) = q$, where $0 < q < 1$

Percentiles are a special case of quantiles, with $q = p/100$

25th percentile: Q_1 . 50th percentile: Q_2 or median

The 25th percentile is the smallest value x such that the cumulative probability (CDF) $F(x)$ reaches or exceeds 0.25.

Let X be a discrete rv with set of possible values S and probability mass function $p(x)$. The **expectation** of X is

$$E[X] = \sum_{x \in S} x P(X = x) = \sum_{x \in S} x p(x).$$

Let X be a random variable with mean $E[X] = \mu < \infty$. The **variance** of X is

$$V[X] = E[(X - \mu)^2] = \sum_x (x - \mu)^2 P(X = x).$$

The **standard deviation** of X is

$$SD[X] = \sqrt{V[X]}.$$

A useful computational formula for the variance is

$$V[X] = E[X^2] - (E[X])^2.$$

10.1

Markov's Inequality

Let X be a non-negative random variable with mean $E[X]$. Then for any positive real number $a > 0$.

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Imagine:

- $E[X] = 5$
- $a = 10$

Then $P(X \geq 10) \leq 5/10 = 0.5$.

So, no matter what the distribution is,

there's **no way** more than half of the probability mass can lie beyond 10.

Chebyshev's Inequality:

Let X be a random variable with finite mean μ and finite variance (σ^2) . Then for any positive real number $k > 0$,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

This inequality provides an upper bound on the probability that X deviates from its mean by more than k standard deviations.

Discrete Distribution

Bernoulli

any random variable that takes exactly two possible values (labeled as 0 and 1) is called a Bernoulli random variable.

We denote this distribution by $\text{Ber}(p)$

A Bernoulli trial is a random experiment consisting of **exactly one trial** with two possible

categorical outcomes, commonly labeled as success (1) and failure (0).

Let X be a Ber RV

Probability mass function:

$$P(X = x) = p(x) = \begin{cases} p^x \cdot q^{1-x}, & \text{for } x = 0, 1 \\ 0 & \text{otherwise.} \end{cases}$$

•

• $E[x] = p \quad V[x] = pq$

Binomial

Consists independent and identical **n trials**

Dichotomous outcomes (success and failure), probability of success p is constant from trial to trial

X=number of successes n = number of trials p= probability of success on one trial

$X \sim \text{Bin}(n, p)$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, 2, \dots, n.$$

• $E[x] = np$ and $V[x] = np(1-p)$

Poisson

Pois (λ)

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

X = the time of occurrence in a given time

Λ = average time of occurrence in a given time

- Models the number of events in **a fixed interval of time or space**.

Hypergeometric

Without replacement / dependent

$$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

N = total number, K = number of samples of that k, n = number of examined trials

$$E[x] = n \frac{K}{N}$$

$$\text{Var}[x] = n \frac{K(N-K)(K-n)}{N(N-1)}$$

continuous random variable: any value within an interval

week 7

Geometric Distribution: A sequence of Bernoulli trials that continue until the **first** success is observed.

X = the number of trials until the first success(inclusive

- n is not fixed
- success and failure
- independent trials
- probability of p is constant

$$P(X=k) = (1-p)^{k-1} p, \quad k=1,2,3, \dots$$

Mean: $E(x) = 1/p$

$$\text{Variance: } \text{var}(x) = \frac{1-p}{p^2}$$

Negative Binomial

A sequence of Bernoulli until when the r-th success occurs (fixed number of success)

X = number of trials needed to obtain r successes

$$\Pr(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \dots$$

Among the first x-1 trials, we need exactly r-1 successes

$$E[x] = \frac{r}{p}$$

$$\text{Var}[x] = \frac{r(1-p)}{p^2}$$

Continuous Distribution

Uniform:

X – Uniform(a, b)

Uniformly distributed on [a,b]

Exponential:

The time between two occurrence of a poisson

Rate λ

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$E[X] = \frac{1}{\lambda} \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

Normal:

Normally distributed with mean and variance

Gamma

The waiting time until r-th occurrence in Poisson

Week 11 Bivariate Distributions

Given two discrete RV X and Y, the joint probability mass function computes the probability mass for $p(x,y)=P(X=x, Y=y)=P(X=x \cap Y=y)$.

Any function that satisfies properties is a valid probability mass/density function.

	Discrete	Continuous
1.	$0 \leq p(x, y) \leq 1$	$f(x, y) \geq 0$
2.	$\sum_x \sum_y p(x, y) = 1$	$\int_x \int_y f(x, y) dy dx = 1$

Bayes:

$$\Pr(A | B) = \frac{\Pr(A) \Pr(B | A)}{\Pr(B)}$$

probability axioms

- **Non-negativity**

$$P(A) \geq 0$$

- **Total probability = 1**

$$P(S) = 1$$

- **Additivity (for mutually exclusive events)**

If events cannot happen together:

$$P(A \cap B) = 0$$

then

$$P(A \cup B) = P(A) + P(B)$$